EVEN ODDS

A fair, verifiable lottery for oversubscribed programs

Camps, swim lessons and community-garden plots, without the 9:00 a.m. rush. Families apply any time during the window, see honest odds, and get a draw that keeps siblings together and that anyone can replay.

Event	Appwrite Databases Hackathon (hackathon.appwrite.io), solo, Oct 19–31, 2026
Deadline	Oct 31, 2026, 11:59 PM PT. Repo + 2–3 min video + optional live link
Database	Appwrite TablesDB (one database type, as the rules require)
Status	Idea locked Oct 24, 2026, after 3 research rounds and a gap review
Document	Design brief: research, design, verified platform limits, gap register, plan

1. The hackathon at a glance

Criterion (unweighted)	What it means for us
Idea and innovation	Original approach to a real problem. Avoid themes already taken by public entries.
Technical execution	Working, complete, deployed. Past Appwrite winners were finished products.
Design and experience	Intuitive and accessible; a judge must be able to try it in 30 seconds.
Usage of Appwrite Databases	"How deliberately the project uses the database you picked." The database feature has to be the product, not storage.

Rules that shape the build: solo only; all code written after Oct 19; open-source libraries allowed if listed; projects must be built on one of TablesDB, DocumentsDB, VectorsDB, Managed PostgreSQL or Managed MySQL.

What Appwrite just shipped (Appwrite 2.0, Aug 31 – Sep 2, 2026)

- Five database types; TablesDB gained geo columns (point/line/polygon), server-side Operators, transactions, offline sync and realtime queries.
- VectorsDB (built-in embeddings, HNSW) and DocumentsDB (schemaless JSON) launched Sep 2.
- Also: Firewall, OAuth2 server, S3-compatible Storage API, MCP events (Oct 19), interval functions (Oct 12).
- Main rival: Supabase (Select conference Oct 2; agents and MCP). Convex owns reactive TypeScript; Firebase owns mobile offline.

The field so far (public GitHub, Oct 24)

About 140 public repos created since Oct 19 mention Appwrite; about 20 are clearly hackathon entries (28 mention TablesDB, 8 VectorsDB, 0 DocumentsDB). Many entrants keep repos private until they submit. Themes already taken: agent memory (Déjà, Engram), neighbourhood help maps (Hearth), tool lending with race-safe claims (LoanLoop), disaster incident claiming (Beacon), volunteer shift boards (SyncBoard), semantic photo search (WeyAm), embedding word games (Almost), incident replay (Rewind).

2. How we chose Even Odds

Three rounds, each one a tournament: research candidates, cut weak or taken ones, score finalists on the four criteria, then check the survivors against existing tools.

Round	Candidates	Winner	Why it lost or won
1. Broad research	24 problems from 3 research agents	Roll Call (missing-person reconciliation, VectorsDB)	Strongest permissions story; but synthetic data, and a disaster theme already crowded by public entries
2. First principles	15 more, in 3 under-explored classes	Even Odds (TablesDB)	Beat Roll Call narrowly: real data, judges can try it, uses TablesDB 2.0 features for their real purpose
Past winners check	Hacktoberfest 2024 top 3	Even Odds scores 33.5 vs 26 / 22 / 20	2024 winners would score low on 2026's database criterion; Algomorph stored its data in NeonDB

Finalist (round 2)	Idea	Tech	Design	DB use	Notes	Overall
Even Odds (TablesDB)	6.5	9	8	10	Rec.us has a basic "fair lottery"; our edge is siblings, replay, odds, zones	33.5
Chorus (VectorsDB)	7	9	8	8	AI comment floods; DocketScope/CommentWorks exist	32.5
Roll Call (VectorsDB)	8	8	8	10	Synthetic data only; disaster theme crowded	32
Recall ledger (DocumentsDB)	6	8	6	9	Same N→M pattern, reads as a dashboard	30

Scores are judgement, not measurement. Overall is the criterion sum adjusted for data realism, overlap and build risk.

3. The problem

> Oversubscribed programs open first-come registration at one moment. The fastest clicker wins, portals buckle, siblings get split across weeks, and nobody can check whether the result was fair.

Evidence	Source
Fairfax County's camp platform "crashed under the strain of first-day registrations"; the county split into zones with staggered openings.	FFXnow, Jan 2025
Bend, Oregon: about 1,200 people in the online waiting room before 6:00 a.m.; about 6,500 enrollments.	Central Oregon Daily, 2026
Chicago Park District camps still sold out within minutes after an April 2026 overhaul.	Block Club Chicago, Apr 2026
Swim registration "like Hunger Games", with "tens of thousands of parents" registering at once.	Richmond News (letter)
Cambridge MA and Girl Scouts of Central Texas run sibling-aware lotteries by hand-written rules; no vendor advertises it.	Official program pages
Scale: an estimated 14–26 million US camp registrations a year across ~14,000 camps (secondary ACA figures).	Secondary sources

Who uses it

Role	What they do
Family	Applies during the window, ranks weeks, chooses a sibling mode, sees banded odds, accepts or declines an offer
Organiser (parks staff)	Loads the catalogue, draws priority zones, verifies residency for offered families, runs the draw
Public / auditor	Sees demand, checks the inputs hash and drand round, replays the draw, reads the fairness report

4. How it works

```
SETUP            APPLY (window)          FREEZE             DRAW                 AFTER
catalog CSV  ->  rank weeks, pick    ->  inputs hash   ->  snake draft over  ->  48h offers
zone polygons    sibling mode; dup       published;        lottery order;       waitlist in
drand round N    children blocked;       overrides         siblings placed      lottery order;
announced        banded hourly odds      locked            per transaction      public replay
```

Step by step

- **Setup.** Organiser imports sessions (week × age group × capacity) and draws priority-zone polygons. The future drand round number N that will seed the draw is published at window open.
- **Apply.** Parents sign in with an email one-time code, add children, rank weeks and request up to K weeks per child. The **child is the ticket**: a unique key on (normalised name, date of birth, program) blocks a second parent from creating a second ticket and links them as a guardian instead.
- **Odds.** Banded, per tier, recomputed hourly by a Monte Carlo function: "chance of at least one week" plus demand per seat. The UI says plainly that ranking honestly never hurts you. The display freezes 24 h before close.
- **Freeze.** At close, all inputs (pseudonymous IDs, tiers, preferences, overrides) are hashed and published in one transaction; the tables are locked by permissions.
- **Draw.** After drand round N is published, the seed is fixed. A pure, deterministic draw module computes the plan; the database enforces it.
- **Offers and waitlist.** 48-hour offers; declines and expiries promote the next family in strict lottery order, inside transactions.
- **Replay.** Anyone can rerun the draw in the browser from the public inputs, verify the drand signature and the inputs hash, and get identical placements.

5. The draw

Random serial dictatorship extended to a multi-week **snake draft**: everyone gets one week before anyone gets a second.

```
seed  = drand(round N)                              # announced in advance, unknowable before
order = families sorted by (tier, sha256(seed || public_id))
for round in 1..K:                                  # K = max weeks per child
    sequence = order if round is odd else reversed(order)
    for family in sequence:
        for week in family.ranked_weeks not yet held:
            seats = ledger rows for (week, each child's age group)
            mode  = together_only | together_preferred | independent
            TRANSACTION (one per family):
                incrementRowColumn(seat_ledger, 'taken', +1, max = capacity) per child
                createRow(placements, child, week)   # unique (child, week)
            on COLUMN_LIMIT_EXCEEDED -> whole family rolls back -> try next week
            on success -> next family
unplaced families join waitlists in lottery order
```

Properties

- **Siblings all-or-nothing** in "together only" mode, because one transaction places the whole family; "together preferred" falls back to per-child placement.
- **No overselling**, guaranteed by the database, not by application code.
- **Round 1 is strategy-proof**: ranking your true favourite first never hurts. Multi-round drafts are not fully strategy-proof; this is disclosed in the deck.
- **Plan in code, enforce in the database.** The same pure module runs on the server and in the browser replay, so they cannot disagree; commits run in parallel batches because the caps and unique index make concurrency safe.
- Age is computed as of each week's start date. Late applicants rank after all on-time applicants with the same hash.

6. Why TablesDB, feature by feature

TablesDB feature	Job in Even Odds
Transactions	Place a sibling group all-or-nothing; release-and-promote a waitlist seat atomically; freeze inputs in one step
incrementRowColumn with max	Seat ledger per (week, age group) that cannot exceed capacity; throws COLUMN_LIMIT_EXCEEDED
Operators (array insert/remove, dateAddDays, toggle)	Reorder ranked weeks in place; server-clock offer expiry; staff deposit flag
Unique indexes	One ticket per child; one placement per (child, week); idempotency per (family, round)
Geo polygons + Query.intersects	Point-in-zone for priority tiers; near-boundary flag for verification
Relationships	Programs → sessions; families ↔ guardians; children → placements
Row-level permissions	Families see only their rows; organisers see tiers, not homes; public sees demand, audit log, pseudonymous inputs
Realtime (with server-side query filters)	Live demand row, instant offers, moving waitlist positions, within the 250-connection Free limit
Bulk operations + CSV import	Load the catalogue; seed 2,000 demo families in ~20 calls

Data model

```
programs     id, org_team, name, window_open, window_close, max_weeks, drand_round, inputs_hash, status
seat_ledger  id, program_id, week, age_group, capacity, taken                    (capped counter)
zones        id, program_id, name, tier, area(polygon)
families     id, public_id, claimed_tier, verified_tier, sibling_mode, home(point, private)
guardians    id, user_id, family_id                                    (many-to-many)
children     id, family_id, first_name, birthdate, child_key  UNIQUE(program, child_key)
preferences  child/family ranked_weeks[array], requested_weeks
placements   id, child_id, week, status, expires_at           UNIQUE(child_id, week)
waitlist     id, program_id, week, family_id, position
overrides    id, family_id, reason_code, pool, created_before_freeze
draw_runs    id, program_id, cursor, inputs_hash, drand_round, status
audit_log    id, program_id, event, payload_hash, prev_hash             (public, append-only)
```

7. Verified platform facts (from Appwrite source and docs)

Design-breaking finding: `Operator.increment(1, max)` inside updateRow **does nothing and reports success** once the cap would be exceeded, so a sold-out seat looks booked. The dedicated `incrementRowColumn(max)` endpoint locks the row and **throws** ("Attribute value exceeds maximum limit"); staged in a transaction, that rolls back the whole family. Confirmed in utopia-php/database 8.3.15, which also ships a CounterRangeRaceTest for this race.

Fact	Value	Design consequence
Values returned while staging in a transaction	Mock values, not real	Never trust them; re-read counters after commit
Transaction conflicts	Timestamp-based; 409 TRANSACTION_CONFLICT; unique violations also 409 + rollback	Retry with jitter; idempotency row per (family, round)
Operations per transaction	Free 100 · Pro 1,000	A family is ~12 operations
Transaction TTL	Default 300 s; 60–3,600 s	Short TTL per family
Interval functions	Free: every 60 min or longer	Hourly odds and expiry sweeps; draw runs on demand
Function timeouts	Sync calls capped at 30 s	Async, resumable draw with a stored cursor
Realtime	Free: 250 connections, 2M messages; server-side filters	Families subscribe to own rows + one aggregate row
Quotas	1 database, 500K reads, 250K writes / month; overage freezes the project	Load test in a scratch project; judges get a clean one
Rate limits	Client SDKs only, not API keys	Run load tests with the server key
Geo	Query.intersects(col, [lng, lat]) is point-in-polygon; boundary counts as inside	Use for tiers; flag homes within 50 m
Bulk	Free 100 rows per request, atomic	Seed in batches
Compare-and-set	None; updateRows with queries does not lock	Rely on capped increments + unique indexes
Email login	createEmailToken (6-digit code)	Passwordless families

Unverified, to test on day 1: Cloud build parity with main; outbound HTTPS from functions (drand is verified in the browser regardless); geo on the Free plan; how transaction internals count against the write quota.

8. Gap register and fixes

Technical

#	Gap	Fix	Priority
T1	Capped operator silently no-ops at the cap	incrementRowColumn(max) in a transaction + unique placement index + post-commit re-read	P0

#	Gap	Fix	Priority
T2	Optimistic conflicts measured from staging time	Jittered retries; unique idempotency row per (family, round)	P0
T3	100 operations per transaction on Free	One transaction per family (~12 ops)	P0
T4	2,000-family draw too slow for one function run	Pure plan, parallel enforced commits, resumable cursor	P0
T5	Interval functions at most hourly	Hourly odds and sweeps; on-demand draw and close	P0
T6	250 realtime connections	Server-filtered subscriptions; server-side load test	P0
T7	Quota overage freezes the project	Separate scratch and judge projects	P0
T8	Outbound HTTPS from functions unverified	Verify drand signature in the browser with drand's JS client	P0
T9	Point-in-polygon method	Query.intersects; boundary flag	P1
T10	Auth flow	createEmailToken OTP; Teams for organisers	P0

Domain (evidence from real lotteries and matching research)

#	Gap	Evidence	Fix	Pri.
D1	Families want several weeks	Gilsland Farm caps 3 weeks/child; SJMA gives one week first	Snake draft with K weeks per child	P0
D2	Operator can re-run until it likes the result	Franklin Towne (2023 allegations); DV-2012 lottery voided	Freeze → inputs hash → pre-announced future drand round → reveal; replay + bias test	P0
D3	Duplicate tickets (two parents, many accounts)	Housing Connect rules; MA audit: 2,342 probable duplicates	Child is the ticket: unique child_key; guardians many-to-many; staff conflict queue	P0
D4	Siblings span age groups; strict all-or-nothing hurts big families	Dur, Morrill & Phan 2022; Correa et al. 2022	Ledger per (week, age group); together-only / preferred / independent	P0
D5	Live odds invite gaming a strategy-proof draw	Boston 2005; arXiv 2501.04243	Banded hourly odds per tier; "at least one week"; honesty message; 24 h freeze	P0
D6	Residency fraud from typed addresses	Housing Connect; Cambridge eligibility review	Claimed vs verified tier; verify offered families only; deterministic demotion; boundary flag	P1
D7	Overrides hide rigging; aides need capacity	Franklin Towne allegations	Pre-freeze overrides with reason codes, hashed and counted publicly; separate accommodation pool	P1
D8	Offer and waitlist races	Girl Scouts deposit model	Transactions + capped increments; strict lottery-order promotion; deposit toggle	P0
D9	Age cutoffs, late and withdrawn applications	Cambridge "Waiting Pool"	Age per week start; late after on-time; withdrawal = decline event	P0
D10	Accessibility, children's data, public records	ADA Title II (WCAG 2.1 AA); amended COPPA (Apr 2026)	H(seed || receipt_secret) IDs; text-table replay; reduced motion; 90-day PII deletion	P2
D11	Multi-round draft not fully strategy-proof	Known result	Round 1 strategy-proof; disclose	P2

Hackathon-specific

#	Gap	Fix	Priority
H1	Judges cannot wait a week for a draw	Judge sandbox: seeded program, apply as a family, fast-forward buttons, per-visitor reset	P0
H2	Database work is invisible	Invariant inspector panel showing each guarantee as it fires	P0
H3	Organisers already use ActiveNet or Rec.us	A lottery layer, not a registration system: embed widget + CSV export	P1

#	Gap	Fix	Priority
H4	"Fair" is just a claim	Fairness report card: placement rate by family size, tier and application time; bias test	P1

9. Selling points and the demo

- **"Try to rig it."** Frozen inputs hash, a future drand round arriving live, a self-verifying replay and a bias test, told alongside the voided 2012 green-card lottery.
- **"Two parents, one ticket."** The second parent's application is blocked by the unique index on screen and linked to the existing child.
- **"The database refused."** The inspector shows a seat cap firing and a sibling group rolling back as a unit.

Time	On screen
0:00	Bend's 6 a.m. waiting room. "Registration rewards the fastest clicker."
0:20	A parent applies in 40 seconds: two kids, three weeks ranked, "together only". Banded odds: at least one week ~80%.
0:45	The second parent tries to apply for the same child. Blocked, then linked as a guardian.
1:00	Organiser view: demand heatmap fills as 2,000 simulated families apply. Freeze: inputs hash published.
1:15	drand round N arrives; draw runs. Inspector: Week 3 / Ages 6–8 → COLUMN_LIMIT_EXCEEDED → rolled back → Week 5 placed.
1:40	Fairness report: 0 oversold, 0 sibling splits in together-only mode, flat placement rate vs application time; first-come comparison on the same data.
2:00	Hammer test: 500 simultaneous accepts for one freed seat; exactly one winner.
2:15	A judge clicks Replay: drand signature verified, inputs hash verified, every placement matches. Embed widget on a parks page.

All numbers in the video will come from real runs; none are invented.

10. Architecture and plan

- **Frontend:** React + Vite + TypeScript, Leaflet maps, deployed on Appwrite Sites; embeddable script-tag widget.
- **Auth:** email one-time codes for families; a Team for organisers.
- **Functions (Node):** draw (async, resumable), odds (hourly), expire-offers (hourly), freeze/close.
- **Shared:** one pure TypeScript draw module used by the server and the browser replay.
- **Data:** Fairfax public camp catalogue and county zone maps, plus 2,000 synthetic families labelled as synthetic.

Day	Date	Work
1	Oct 24–25	Live checks on Cloud (capped increment throws and rolls back; Query.intersects; realtime filters; function timeout/async), schema, seed data
2	Oct 25	Pure draw module (snake draft, sibling modes, tiers), replay parity tests, commit layer
3	Oct 26	Apply flow with duplicate blocking, banded odds, zone tiers
4	Oct 27	Freeze timeline, drand, offers and waitlist, judge sandbox
5	Oct 28	Inspector, fairness report, load test, widget and CSV export
6	Oct 29	Polish, accessibility pass, deploy, README, video
7	Oct 30–31	Buffer; submit before Oct 31, 11:59 PM PT

Fallback

If the capped increment does not throw and roll back inside transactions on Cloud, and the unique-index fallback also fails, switch to Roll Call (VectorsDB), which needs neither.

Honest limits

- No payments: the flow ends at an accepted offer (staff can mark a deposit).
- US geocoding only; demo families are synthetic.
- A lottery is not automatically fair to need: tiers and accommodation pools can encode policy, but setting it is the organiser's call.
- Rec.us already advertises a basic fair lottery; Even Odds' differentiators are sibling atomicity, verifiable replay, honest odds and zone tiers.

Sources

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- drand public randomness beacon: drand.love